

Open Science in Practice: Collaborative Research with Git



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K8F6106
HT2026



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Positron and environment management

Niklas Edvall

Rterm (64-bit)

Microsoft Windows [Version 10.0.19045.6456]
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C:\Users\nikedv>R

R version 4.3.2 (2023-10-31 ucrt) -- "Eye Holes"
Copyright (C) 2023 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
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R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> a <- 13
> a < 14
[1] TRUE
>

EnvironmentHistoryConnectionsTutorial

Global Environment

Values

age	num	[1:120]	49	50	47	61	23	16	26	3...		
n			120									
sex	num	[1:120]	1	2	1	1	1	1	2	1	1	...

Untitled1 × simulate_dataset_1.R

```
1
2 # Simulate lifestyle + mental health dataset
3 n <- 120
4
5 set.seed(2025)
6
7 # Sex: 1 = male, 2 = female, 3 = other
8 # Mostly binary with a few "other"
9 sex <- sample(
10   x = c(1, 2, 3),
11   size = n,
12   replace = TRUE,
13   prob = c(0.48, 0.48, 0.04)
14 )
15
16 # Age: normal around 40, truncated to 16-80
17 age <- round(rnorm(n, mean = 40, sd = 15))
18 age <- pmin(pmax(age, 16), 80)
19
20 # How did you hear about the survey?
21 survey <- sample(
22   x = 1:4,
23   size = n,
24   replace = TRUE,
25   prob = c(0.05, 0.60, 0.15, 0.20)
26 )
27
28 # Set 3 random observations to missing
29 missing_indices <- sample(1:n, size = 3)
30 survey[missing_indices] <- NA
31
32 # -----
33 # Position
34 # 1 = student
35 # Everyone ≤25 is position 1
```

FilesPlotsPackagesHelpViewerPresentation

ConsoleTerminalBackground Jobs

```
> R 4.3.2 ~/  
>  
> # Sex: 1 = male, 2 = female, 3 = other  
> # Mostly binary with a few "other"  
> sex <- sample(  
+   x = c(1, 2, 3),  
+   size = n,  
+   replace = TRUE,  
+   prob = c(0.48, 0.48, 0.04)  
+ )  
>  
> # Age: normal around 40, truncated to 16-80  
> age <- round(rnorm(n, mean = 40, sd = 15))  
> age <- pmin(pmax(age, 16), 80)  
> plot(age,sex)  
> plot(sex, age)  
> bar(sex, age)  
  
Error in bar(sex, age) : could not find function "bar"  
  
> boxplot(sex, age)  
> |
```


spyder

plugin.py

- Plots
 - get_name
 - get_description
 - get_icon
 - register
 - unregister
 - current_widget
 - add_shellwidget
 - remove_shellwidget
 - set_shellwidget
- chart_plot_example.py
 - Plot final terrain model
 - generate_polar_plot
 - generate_dem_plot
 - main
- plugin.py
 - IPythonConsole
 - _init_
 - SpyderPluginMixin API
 - update_font
 - _apply_gui_plugin_settings
 - _apply_mpl_plugin_setting
 - _apply_advanced_plugin_s
 - _apply_pdb_plugin_setting
 - apply_plugin_settings_to_c
 - apply_plugin_settings
 - toggle_view
 - SpyderPluginWidget AP
 - get_plugin_title
 - get_plugin_icon
 - get_focus_widget
 - closing_plugin
 - refresh_plugin
 - get_plugin_actions
 - register_plugin

```
1  -*- coding: utf-8 -*-
2  #
3  # Copyright © Spyder Project Contributors
4  # Licensed under the terms of the MIT License
5  # (see spyder/_init_.py for details)
6
7  """
8  Plots Plugin.
9  """
10
11 # Third party imports
12 from qtpy.QtCore import Signal
13
14 # Local imports
15 from spyder.api.plugins import Plugins, SpyderDockablePlugin
16 from spyder.api.translations import get_translation
17 from spyder.plugins.plots.widgets.main_widget import PlotsWidget
18
19 # Localization
20 _ = get_translation('spyder')
21
22
23 class Plots(SpyderDockablePlugin):
24     """
25     Plots plugin.
26     """
27     NAME = 'plots'
28     REQUIRES = [Plugins.IPythonConsole]
29     TABIFY = [Plugins.VariableExplorer, Plugins.Help]
30     WIDGET_CLASS = PlotsWidget
31     CONF_SECTION = NAME
32     CONF_FILE = False
33     DISABLE_ACTIONS_WHEN_HIDDEN = False
34
35     # --- SpyderDockablePlugin API
36
37     def get_name(self):
38         return _('Plots')
39
40     def get_description(self):
41         return _('Display, explore and save console generated plots.')
42
43     def get_icon(self):
44         return self.create_icon('hist')
45
46     def register(self):
47         # Plugins
48         ipyconsole = self.get_plugin(Plugins.IPythonConsole)
49
50         # Signals
51         ipyconsole.sig_shellwidget_changed.connect(self.set_shellwidget)
52         ipyconsole.sig_shellwidget_process_started.connect(
53             self.add_shellwidget)
54         ipyconsole.sig_shellwidget_process_finished.connect(
55             self.remove_shellwidget)
56
```

Name	Type	Size	Value
bool	bool	1	True
data	Array of str128	(3, 3)	ndarray object of numpy module
datetime_object	datetime	1	2021-04-14 17:35:14.687085
df	DataFrame	(2, 2)	Column names: Col1, Col2
filename	str	53	/Users/Documents/spyder/spyder/tests.test_dont_use.py
li	list	5	['abcd', 745, 2.23, 'efgh', 70.2]
myset	set	3	{'2', '1', '3'}
r	float	1	6.46567886443
t	tuple	5	('abcd', 745, 2.23, 'efgh', 70.2)
tinylist	list	2	[123, 'efgh']

conda: spyder-dev (Python 3.8.5) LSP Python: ready master Line 10, Col 1 UTF-8 LF RW Mem 64%

The open source AI code editor

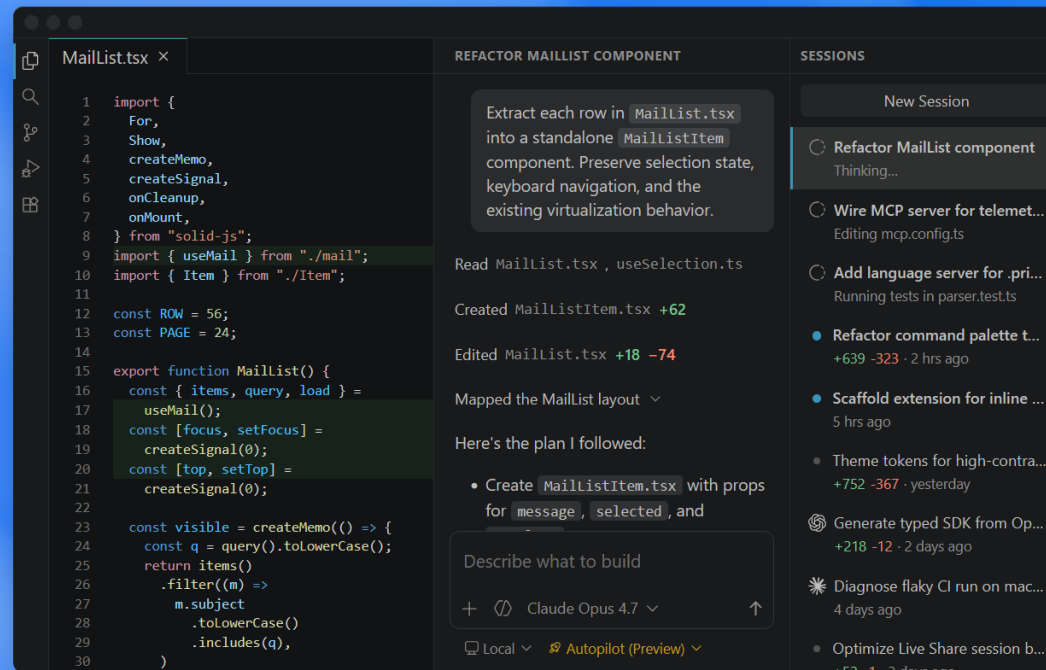
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Fork 42.4k

Star 193k

main

5189 Branches

391 Tags

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Benjamin Christopher Simmonds (benibenj)

Merge pull request #335957 from ...

f80869a · 1 hour ago

165,378 Commits

.agents/skills/launch	sessions: streamline repository selection (#335358)	last week
.config	chore: suppress node warnings (#274905)	11 months ago
.devcontainer	Add sshd to devcontainer features (#322971)	2 months ago
.eslint-plugin-local	eslint: allow bracket notation for process.env (#335596)	5 days ago

About

Visual Studio Code

code.visualstudio.com

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typescript

visual-studio-code

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Security

Visual Studio Code - Open Source ("Code - OSS")

feature-request issues 4k open

bug issues 6.1k open

The Repository

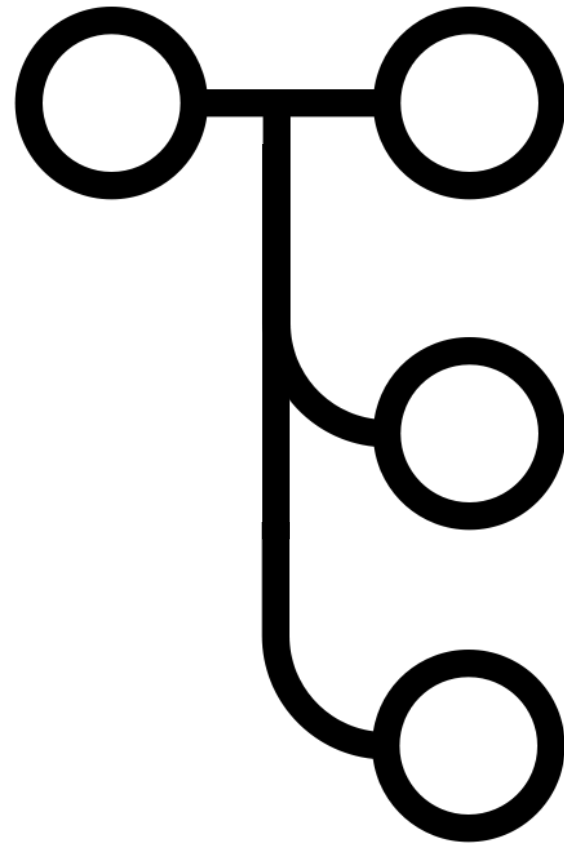
This repository ("Code - OSS") is where we (Microsoft) develop the Visual Studio Code product together with the community. Not only do we work on code and issues here, but we also publish our [roadmap](#), [monthly iteration plans](#), and our [endgame plans](#). This source code is available to everyone under the standard [MIT license](#).

Visual Studio Code

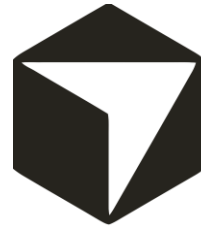
File Edit Selection View Go Run ... demo-travel-blog

EXPLORER JS theme.js 1.1M

CHAT CODEX



~~W Windsurf~~



CURSOR



VSCodium

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Top 20 Best VSCode Extensions for 2026

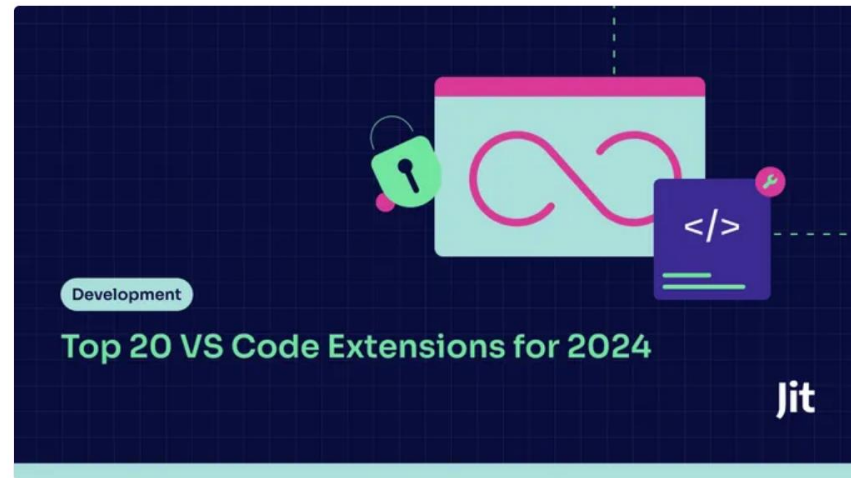


By Eyal Katz



Edited by Jit Team

Updated August 19, 2025



An IDE is like your home. When you first move into a new apartment, it doesn't feel quite right until you've made it yours. IDEs are the same; changing the color scheme, code formatter, and other customization makes it more personalized and helps increase your productivity.

With nearly 75% of the market share and 14 million users, **VSCode is the most popular IDE available.** And because it's one of the best VSCode



Demo

Packages and environments in R and Python



renv – R package that can take and save snapshot of your R project.

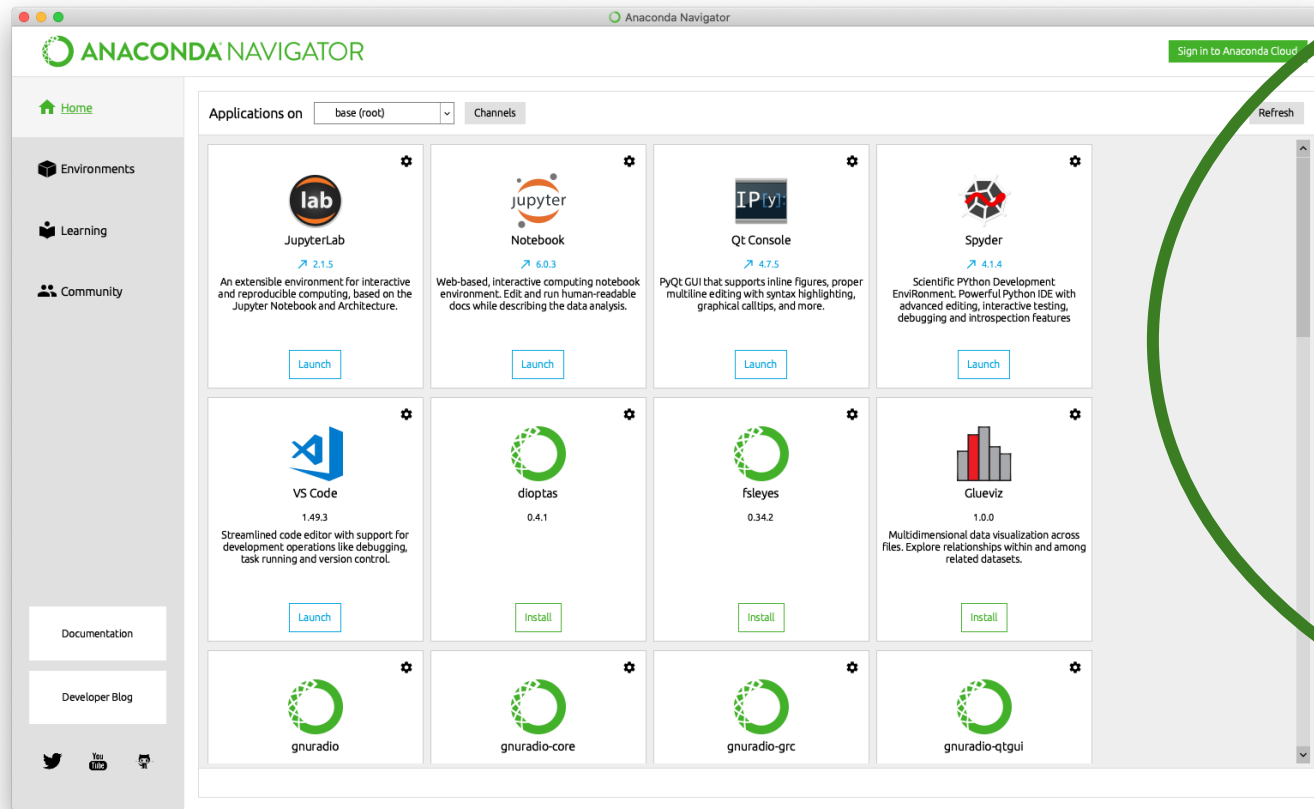
renv::init() → **renv::snapshot()** → **renv::restore()**



venv – Create environment for Python → **requirements.txt**

pip – Install python packages, e.g **pip install requirements.txt**

Packages and environments



ANACONDA®

Graphical user interface
~ 150 pre-installed modules (e.g. python)

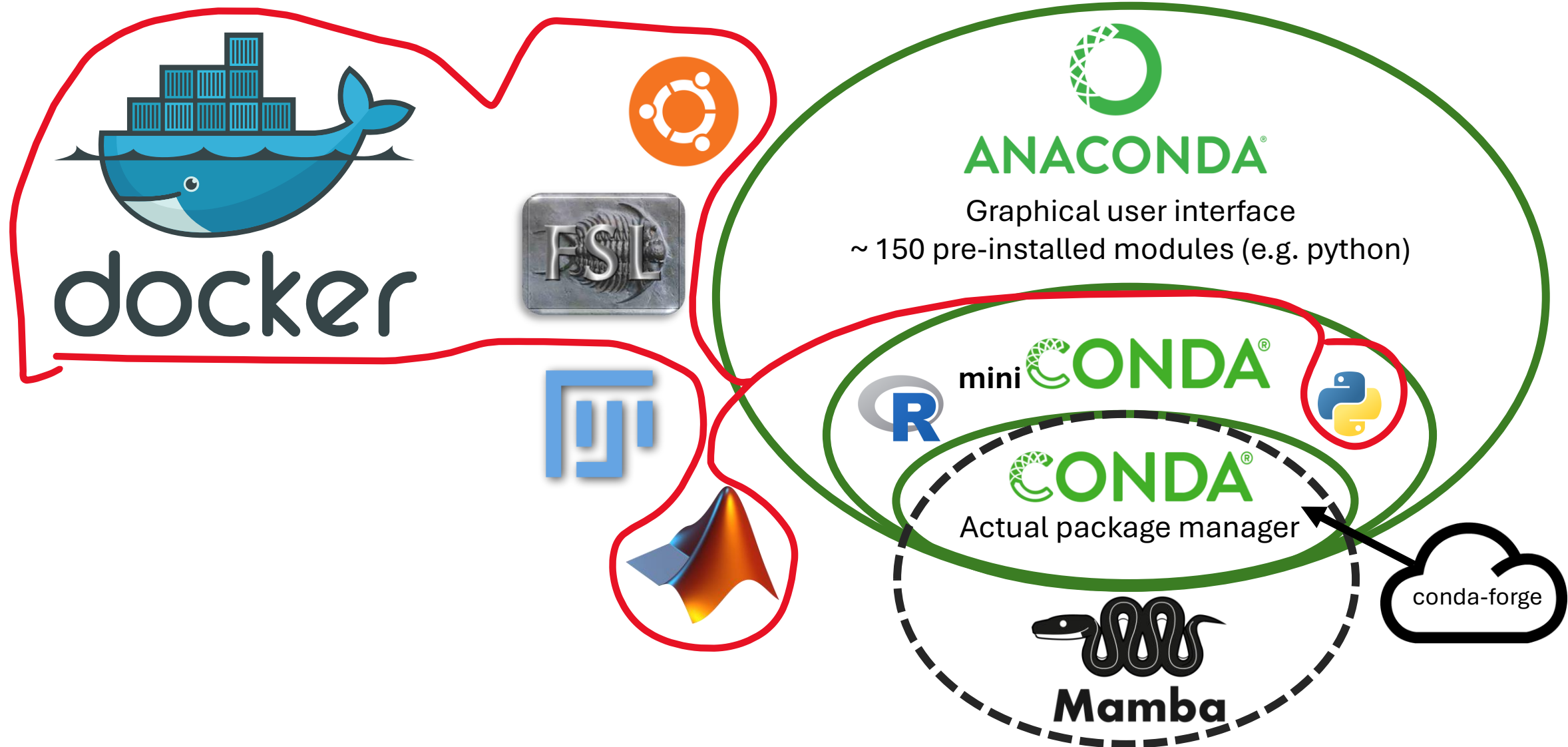


CONDA®

Actual package manager

conda-forge

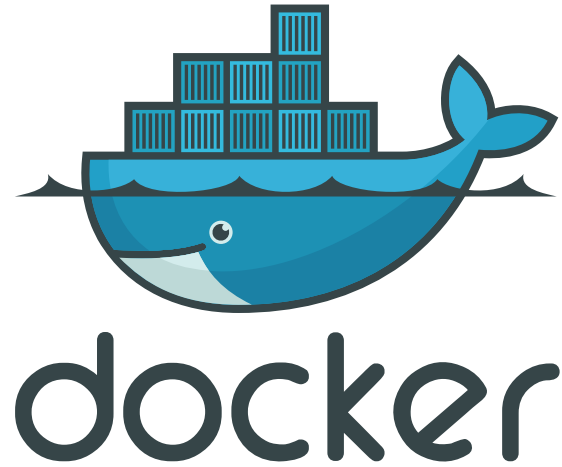
Packages and environments



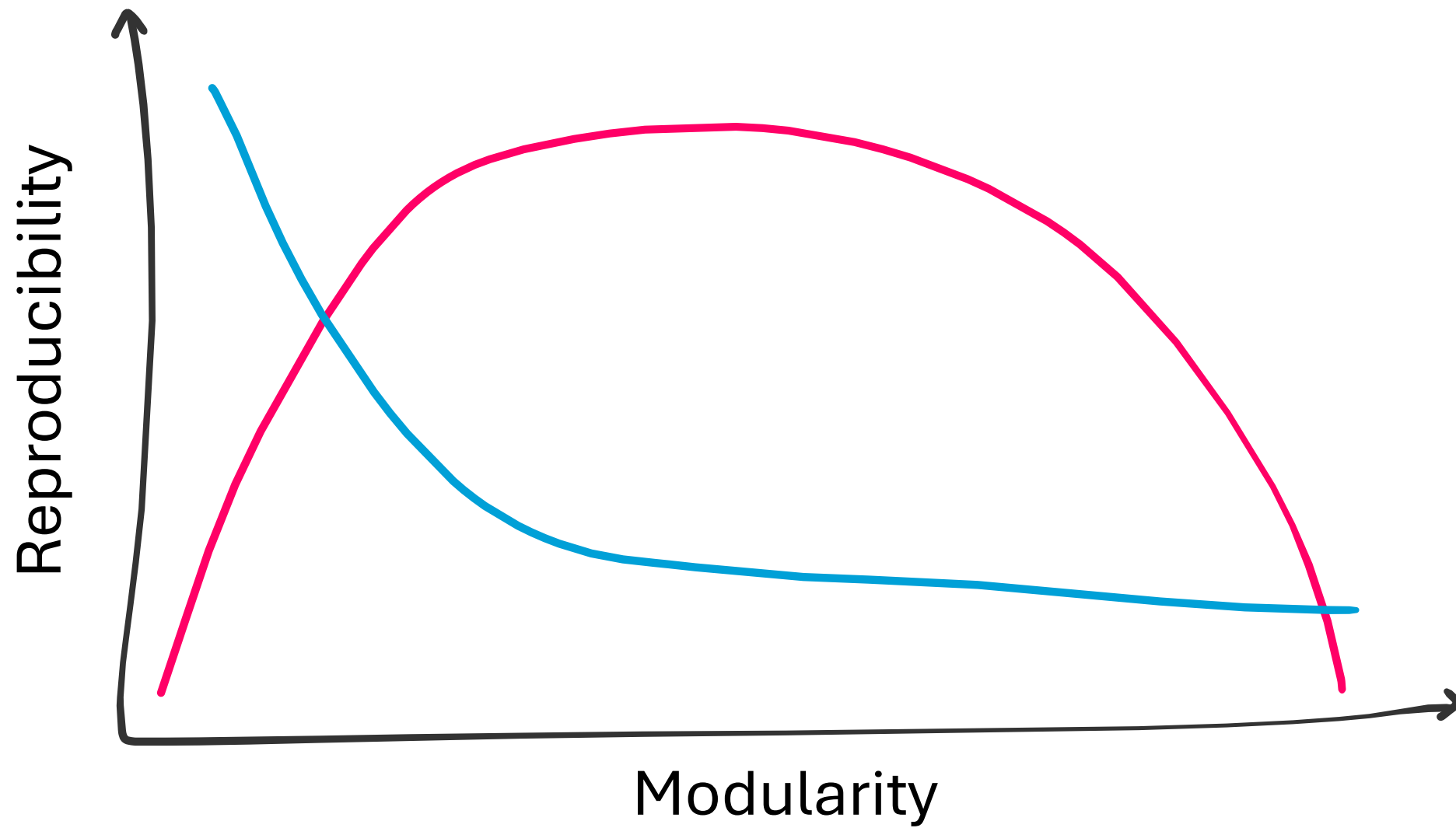
Packages and environments

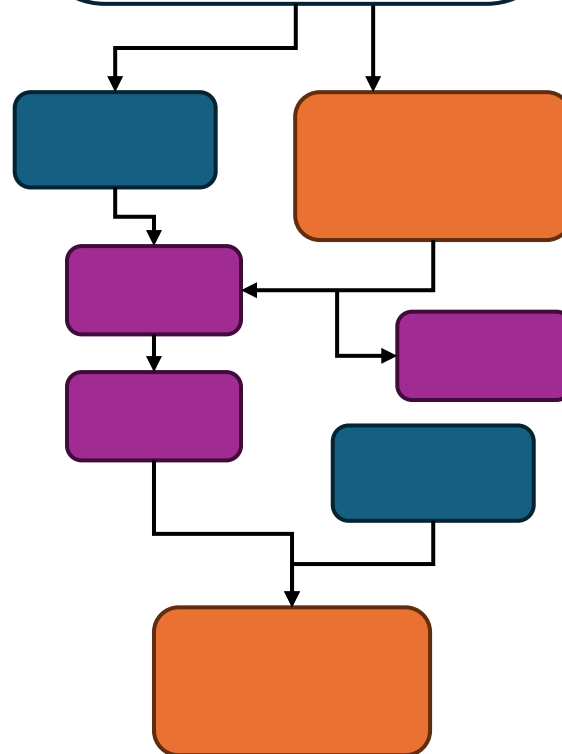
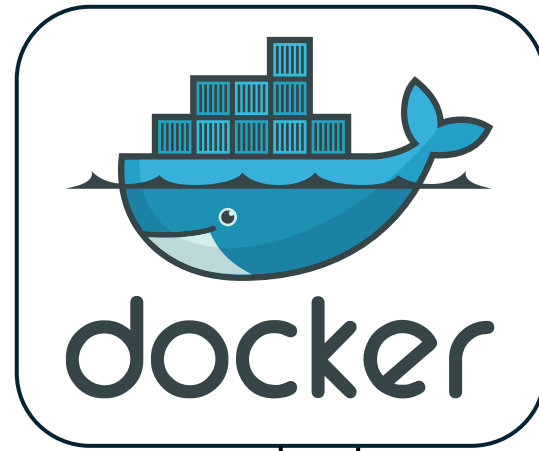


Packages and environments



Apptainer/Singularity





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